

A-3 Discrete Moments

0. Calculator TI-30XS MultiView Overview

1. Mode

- There are 3 common ways to describe the value of a random variable.
 - Mode $\rightarrow$ most common value
 - Median $\rightarrow$ middle value
 - mean $\rightarrow$ avg. value.
- X can be defined as a random variable if it is a no. that depends on choice.
 - $X: S \rightarrow R$, $S =$ sample space
 - $\rightarrow$ it is discrete RV if we can list all its possible values.

$$\boxed{\sum_x P(X=x) = 1}$$

for any discrete RV.

- For a discrete RV X , y is the mode of X if, $\boxed{P(X=y) \geq P(X=x) \forall x}$

Mode y is that input which maximizes $P(X=y)$.

If it is not unique, then X can have multiple modes.

Ex. 7-sided die $\Rightarrow 1, 1, 1, 2, 4, 4, 6$

Mode $\Rightarrow P(X=1) = \frac{3}{7}, P(X=2) = \frac{1}{7}$
 $\swarrow$
 $P(X=4) = \frac{2}{7}, P(X=6) = \frac{1}{7}$

Ex. mode of poisson RV with mean 2.8.

$$P(X=x) = e^{-\lambda} \times \frac{\lambda^x}{x!}$$
$$= e^{-2.8} \times \frac{2.8^x}{x!} \Rightarrow x=2$$

Now, put values of x from 0 to some large no., that that value for which $P(X=x)$ is highest.

2. Median

- CDF of X (Cumulative distribution funcⁿ)

$$F(x) = P(X \leq x)$$

- The median of X is the smallest m s.t.

$$P(X \leq m) = F(m) \geq \frac{1}{2}$$

Eq. 7-sided die $\Rightarrow 1, 1, 1, 2, 4, 4, 6$
 $P(X \leq 1) = \frac{3}{7}$, $P(X \leq 2) = \frac{4}{7} \geq \frac{1}{2}$
 Median = 2

- The 100% p^{th} percentile π_p is the smallest possible x s.t.

$$P(X \leq x) \geq p$$

$x =$	0	1	2	4	6
$F(x)$	0	$3/7$	$4/7$	$6/7$	1

10th $p \Rightarrow P(X \leq 1) \geq 0.1 \Rightarrow 1$

Eq. What is the median of fair 6-sided die

$$P(X \leq 3) = \frac{1}{2}, \quad P(X \leq 4) = \frac{2}{3}$$

so anything $\Rightarrow 3 \leq x < 4 \Rightarrow$ is a median

Eq. $P(N=n) = n/15$ for $1, 2, 3, 4, 5$. Median?

$$P(X \leq 1) = \frac{1}{15}, \quad P(X \leq 2) = \frac{3}{15}, \quad P(X \leq 3) = \frac{6}{15}$$

$$P(X \leq 4) = \frac{10}{15} \checkmark$$

median

3. Mean: definition

eg. 10 people. Owe \$4 to 2, \$2 to 1, & \$1 to 2 people. What is avg. debt?

$$\text{avg. debt} = \frac{\text{total amt.}}{\text{total people}} = \frac{12}{10} = 1.2$$

$$E[X] = \sum_x x \cdot P(X=x)$$

$$= \frac{4 \times 2}{10} + \frac{2 \times 1}{10} + \frac{1 \times 2}{10} = 1.2$$

- If X is a discrete RV, then

$$E[X] = \sum_x x \cdot P(X=x)$$

- You can generalize this concept,

$$E[X^2] = \sum_x x^2 \cdot P(X=x)$$

$$E[g(X)] = \sum_x g(x) \cdot P(X=x)$$

eg. $P(X=x) = \begin{cases} \frac{6-k}{15} & k=1 \text{ to } 5 \\ 0 & \text{otherwise} \end{cases}$

$$E[g(X)] = 100 \times \frac{5}{15} + \dots + 400 \times \frac{1}{15} = 220$$

Eq. $P(N=n) = n/15, n=1 \text{ to } 5. E[n] = ?$

$$E[n] = \frac{1 \times 1}{15} + \frac{2 \times 2}{15} + \frac{3 \times 3}{15} + \frac{4 \times 4}{15} + \frac{5 \times 5}{15}$$

$$= 3.67$$

- just recall, median & mode must be the possible values of the variable. But, mean or the expected value doesn't have to be so.

4. Mean: Tools

- Law of total expectation $\Rightarrow$ If A_1 to A_n are partitions of a sample space, $S = \bigcup_{i=1}^n A_i$ & $A_i \cap A_j = \emptyset$ for $i \neq j$.

$$E[X] = \sum_{i=1}^n E[X|A_i] \cdot P(A_i)$$

- Linear combinations.

$$E[aX + b] = a \cdot E[X] + b$$

In general,

$$E[aX + bY] = a \cdot E[X] + b \cdot E[Y]$$

Eq. $H \Rightarrow 40\%, 500; L \Rightarrow 60\%, 100$

Cost is $1.05X + 10$

$$E[\text{cost}] = E[1.05X + 10] \\ = 1.05 \cdot E[X] + 10$$

$$E[X] = 0.4 \times 500 + 0.6 \times 100 \\ = 200 + 60 = 260$$

$$E[\text{cost}] = 273 + 10 = 283$$

$$\begin{array}{r} 105 \\ 26 \\ \hline 630 \\ 2100 \\ \hline 2730 \end{array}$$

5. Survival Approach

- If $N \geq 0$, & N is integer valued variable.

$$E[N] = P(N > 0) + P(N > 1) + P(N > 2) \dots$$

$$E[N] = \sum_{n=0}^{\infty} P(N > n)$$

$$k = n+1 \text{ so } P(N > n) = P(N \geq n+1)$$

$$E[N] = \sum_{k=1}^{\infty} P(N \geq k)$$

- $P(N > n) = 1 - F(n) \Rightarrow$ survival funcⁿ.

eg. $P(N \geq k) = \frac{5-k}{5} \quad k=0, \dots, 5$

$$E[N] = \sum_{n=0}^5 P(N > n)$$

$$= \frac{4}{5} + \frac{3}{5} + \frac{2}{5} + \frac{1}{5} + 0 = 2$$

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b. Variance : Definition

- There are many cases, where you will see that the mean of different random variables are same. But, they might differ in spread. For such cases, variance of the RV is calculated.

- $$\boxed{\text{Var}(X) = E[(X_i - \mu_X)^2]} \quad \mu_X = E[X]$$

- $E[X^k]$ is the k^{th} moment of X . Also, c/d as the k^{th} raw moment of X .

$$\mu = E[X] \Rightarrow 1^{\text{st}} \text{ moment}$$
$$E[X^2] \Rightarrow \text{second moment.}$$

$$\text{Var}(X) = E[(X - \mu)^2] = \sigma^2 \Rightarrow 2^{\text{nd}} \text{ central moment of } X.$$

eg. $P(X=k) = \frac{6-k}{15}$, $k = 1 \dots 5$, mean = 220
Var = ?

$$E[X^2] - (E[X])^2 = \dots \Rightarrow (220)^2 = 48400$$

$$E[X^2] = \frac{(100)^2 \cdot 5}{15}$$

$$\text{Var}(X) = 10,600$$

7. Variance: Tools

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}$$

$$\Rightarrow E[X^2] \geq (E[X])^2$$

$$\boxed{\text{Var}(cX) = c^2 \text{Var}(X)}$$

$$\boxed{\text{Var}(X+b) = \text{Var}(X)}$$

$$\text{Var}(aX+b) = \text{Var}(aX) = a^2 \text{Var}(X)$$

$$\boxed{\text{SD}(X) = \sqrt{\text{Var}(X)}}$$

$$\Rightarrow \text{SD}(cX) = |c| \cdot \text{SD}(X)$$

- Coefficient of Variation $\Rightarrow \text{CV}(X)$

$$\boxed{\text{CV}(X) = \frac{\sigma}{\mu} = \frac{\text{SD}(X)}{E(X)}}$$

$$\text{CV}(cX) = \text{CV}(X)$$

Eg. $E[X] = 5$, $\text{SD}(X) = 3$

(a) $E[X^2] \Rightarrow \text{Var}(X) = 9$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = 9 + (5)^2 = 34$$

(b) $\text{Var}(2X+6) = 4 \times 9 = 36$

(c) $\text{CV}(2X+6) = \frac{\text{SD}(2X+6)}{E[2X+6]} = \frac{6}{16} = \frac{3}{8}$

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8. Discrete Uniform

- X is a (discrete) uniform on $\{1, 2, \dots, n\}$ if $P(X=i) = 1/n$, for all those n choices.

then, $\boxed{E[X] = \frac{n+1}{2}}$ and $\boxed{\text{Var}(X) = \frac{n^2-1}{12}}$

Ex. $X \in U\{3, 4, 5, 6, 7, 8\}$. $E[X] = ?$

$$Y = X + 2$$

$$E[Y] = E[X + 2] = E[X] + 2$$

$$E[Y] = \frac{6+1}{2} = 3.5 \rightarrow E[X] = 5.5$$

but variance remains same, even if the shift happens.

$$\text{Var}(Y) = \text{Var}(X) = \frac{35}{12}$$

- In a general case, if X is uniform on $\{a, \dots, b\}$

$$\boxed{E[X] = \frac{a+b}{2}}$$

$$\text{Var}(X) = \frac{(\text{no. of values})^2 - 1}{12}$$